GBD Child-Health Data Pipeline – Final Documentation

Date: 8 July 2025 | Version: 3.3 (post-cleanup)

Objective

Provide a single-command, reproducible pipeline that assembles a complete **GBD Input-Data Package** for child health by: 1. Extracting the latest risk-factor indicators from WHO & World Bank APIs. 2. Standardising country names, pooling duplicate sources and imputing gaps via mortality-based strata. 3. Delivering ready-to-use outputs (CSV + Excel) for downstream GBD modelling.

Data flow

1. `gbd_pipeline_final.py` orchestrates extraction consolidation tidy transformation template population.
2. Raw API responses are cached in RAM only – no persisted scrap files post-run.
3. `gbd_enhanced_template_populator.py` handles imputation rules and creates `GBD_Input_Data_Package.xlsx`.
4. Generated files live in `gbd_processed_data/` and project root. All other transient files are auto-cleaned.

Indicators covered

Category	Indicator	Source
Nutrition	Malnutrition (weight-for-age z<-2)	World Bank
Nutrition	Wasting prevalence (U-5)	WHO-GHO
Birth	Low birth weight (2500 g)	World Bank
Feeding	Exclusive breastfeeding <6 months	WHO-GHO
Household	Solid fuel use	World Bank
Mortality	U5MR	World Bank
Population	Population ages 0-4 (absolute)	UNICEF SDMX

Imputation hierarchy

1. Latest country-specific value (direct).
2. Mortality-based **sub-regional** average.
3. World-Bank **regional** average.
4. **Global** mean.

Usage recap

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
GBD_PIPELINE_AUTOCONFIRM=1 python gbd_pipeline_final.py
```

Outputs: * `gbd_processed_data/gbd_consolidated_data.xlsx` (long) * `gbd_processed_data/country_list.csv` * `gbd_processed_data/gbd_input_data_matrix.csv` (wide) * `GBD_Input_Data_Package.xlsx` – 133 countries, 0 missing risk factors

Limitations & next steps

- Two micro-states (Cook Islands, Niue) lack population data; integrate UN-DESA pop table for full coverage.
- Add CI to rerun monthly and open PR if any indicator changes.

Data sources & citations

Source	Endpoint / Product	Indicators Used	Access Method	Citation
World Bank HNP API	`/country/all/indicator/SH.STA.MALN.ZS` etc.	Malnutrition, Low birth weight, Solid fuel use, Total population, U5MR	`requests` JSON pagination	World Bank. Health Nutrition and Population Statistics. https://data.worldbank.org
WHO GHO OData	`https://ghoapi.azureedge.net/api/{code}`	Wasting prevalence, Exclusive breastfeeding	`requests` with OData `$format=json`	World Health Organization. Global Health Observatory. https://www.who.int/data/gho

Source	Endpoint / Product	Indicators Used	Access Method	Citation
UN IGME (via WB)	Indirectly via WB U5MR series	Under-5 mortality rate (SDG 3.2.1)	Same as WB call above	United Nations Inter-agency Group for Child Mortality Estimation, 2024
UNICEF SDMX API	`https://sdmx.data.unicef.org/ws/public/sdmxapi/rest/data/UNICEF,DM,1.0/.DM_POP_U5`	Population ages 0-4 (absolute counts)	requests (SDMX-JSON)	UNICEF Demography – Population under age 5. https://data.unicef.org
Master template	Provided file `Master Spreadsheet (DO NOT CHANGE YET).xls`	Baseline countries + columns	`xlrd`	—

All data were retrieved on **8 July 2025**.

Mortality-based sub-regional stratification (imputation engine)

1. **Source metric:** Latest available Under-5 Mortality Rate (U5MR) for each country (numeric per 1000 live births).
2. **Region of reference:** Countries are first mapped to one of seven World Bank analytical regions using an extended alias dictionary.
3. **Quartile thresholds:** For every WB region, the 25th (Q1) and 75th (Q3) percentiles of U5MR are computed.
4. **Strata assignment:**
    - Low child-mortality = U5MR < Q1
    - Medium child-mortality = Q1 U5MR Q3
    - High child-mortality = U5MR > Q3 The stratum label is concatenated with the region (e.g. `SSA - High Child Mortality`).

5. **Imputation hierarchy:** When a country–indicator value is missing, the pipeline seeks (in order):
    a. A direct country value (latest year).
    b. The sub-regional mean for that indicator.
    c. The WB regional mean.
    d. Global mean.

This approach preserves intra-regional heterogeneity and produces realistic proxy values, especially for risk factors correlated with child survival.

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